INTERNATIONAL MONETARY FUND

UNION OF COMOROS

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REQUEST FOR DISBURSEMENT UNDER THE RAPID CREDIT FACILITY AND PURCHASE UNDER THE RAPID FINANCING INSTRUMENT—DEBT SUSTAINABILITY ANALYSIS

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Prepared by the staffs of the International Monetary Fund (IMF) and the World Bank (WB)

Union of Comoros Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress	*Moderate*
Overall risk of debt distress	*Moderate*
Granularity in the risk rating	*Limited Space*
Application of judgement	*No*

Comoros remains at moderate risk of external debt distress, but its space to absorb shocks is "limited." All debt burden indicators exhibit a continual upward trend, with the PV of debt-to-export approaching its threshold at the end of the assessment horizon (2029) under the baseline scenario. (Thresholds reflect "medium" capacity to carry debt). The reduced space to absorb shocks reflects the taking on of a large new loan, a downward revision of projected exports in line with lower export prices and impacts of Cyclone Kenneth on debt accumulation. Shock scenarios indicate vulnerability to a deterioration of export performance, natural disasters, and exchange rate instability. Comoros' overall risk of debt distress remains moderate, given that domestic debt is expected to remain minimal. The authorities need to strengthen policies to improve macroeconomic performance including by making faster progress on domestic resource mobilization and broadening the export base. The authorities should proceed cautiously on taking up any new debt and may wish to largely avoid new non-concessional debt.

PUBLIC SECTOR DEBT COVERAGE

1. The coverage of public sector debt remains unchanged from the most recent DSA. The debt stock covers the central government, as well as government-guaranteed debt of state-owned enterprises (SOEs). SOEs cannot access the debt market on their own, nor can the social security fund. External debt is defined using a residency criterion.

Text Table 1. Public Sector Debt Coverage Under the Baseline Scenario

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	
8	Non-guaranteed SOE debt	

Sources: IMF staff.

2. The contingent liability stress test accounts for the likely costly resolution of the insolvent state-owned postal bank (SNPSF) and associated financial market risk. Accounting for 20 percent of the financial sector deposits, SNPSF is a macro-critical financial institution. The contingent liabilities from financial markets account for the cost of its restructuring, which is added to the standard minimum value of 5 percent of GDP that represents the average cost to the government of a financial crisis in LICs. The contingent liability for SOE debt is set to zero, since all the government-guaranteed debt is included in the DSA. The contingent liability of other elements of the general government is set at 1.8 percent of GDP, the estimated level of domestic arrears at end-2018. Overall, Comoros' total contingent liabilities are estimated at 10.6 percent of GDP.

Text Table 2. Coverage of the Contingent Liabilities' Stress Test

1	The country's coverage of public debt	The central government plus social security, government-guaranteed debt		
		Default	Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0 percent of GDP	1.8	
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	2.0	
4	PPP	35 percent of PPP stock	0.0	
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	6.8	
	Total (2+3+4+5) (in percent of GDP)		10.6	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

Sources: IMF staff.

BACKGROUND

Recent Debt Developments

3. Comoros benefited from extensive debt relief at the beginning of this decade. After reaching the HIPC Completion Point in 2012, Comoros' external debt was reduced by 12.7 percentage points of GDP, to 10.9 percent of GDP at end-2013.

4. Comoros' external public and publicly guaranteed (PPG) debt is mainly held by official bilateral creditors, and most is on concessional terms. The debt held by official bilateral creditors more than doubled during 2015-18, increasing its share in total external debt to 60.2 percent. At end-2018, all external debt is on concessional terms, with fixed interest rates, except for a loan from the French export credit insurer COFACE (EUR 3.4 million, 0.3 percent of GDP).

Text Table 3. Nominal Stock of External PPG Debt, 2018[1]
(Millions of U.S. Dollars; end-of-period)

Total External Debt[2]	**224.1**
Multilateral Creditors	**30.3**
IMF	11.0
IDA	19.3
Bilateral Creditors	**145.0**
Saudi Arabia	44.9
India	38.9
China	30.9
Kuwait	25.9
France	3.4
Mauritius [3]	0.9
Others	**48.8**

[1] Following Paris Club cancellation of all its HIPC-eligible debt, rescheduling of short-term debt in arrears, and restructuring non-Paris club debt.
[2] Excludes $2.72mn of hospital debt owed by Comoros to France that is the subject of ongoing negotiations.
[3] The Mauritius loan and its arrears were canceled in early 2019.
Source: Comorian authorities.

5. Comoros has continued to take on concessional debt. Since 2013, Comoros has contracted five new external concessional bilateral loans, totalizing US$158.7 million, to construct a thermal electricity generation plant, rehabilitate the road network, and strengthen telecommunications infrastructure. The newest loan (US$83 million, 7 percent of GDP), contracted with a Chinese bank in late 2018 to strengthen telecommunication infrastructure, has a maturity of 30 years, including a 5-year grace period, and an interest rate of 1 percent, implying a grant element of 44.7 percent. The undisbursed amounts of these loans are included in the DSA baseline scenario on the grounds of disbursement projections provided by the authorities. Domestic borrowing is limited (to 0.1 percent of 2018 GDP) a three-year loan contracted with the local representative of India's EXIMBANK in 2017.

6. The World Bank (WB) recently did not provide a waiver that would have enabled Comoros to take on a large non-concessional loan. In 2018, the authorities contracted from the Eastern and Southern African Trade and Development Bank (TDB) a non-concessional loan of (Euro 40 million, 3.7 percent of GDP). Disbursement of this loan requires a non-concessional borrowing waiver from the WB, which the Bank did not provide. Against this backdrop, the present DSA highlights the implications for debt sustainability of contracting the TDB loan, where the authorities will make full use of US$15 million in 2019.

7. The authorities' action plan for clearing external arrears was not fully executed. This plan envisaged gradually and completely clearing arrears by March 2019 and seeking rescheduling or cancelation of debt and arrears where possible. The latest available information indicates that the authorities did not fully implement this plan, and at end-2018 external arrears reached US$5.9 million (0.5 percent of 2018 GDP). Of these, arrears of US$2 million (0.2 percent of GDP) were owed to France, Kuwait, India, and Saudi Arabia. Another US$0.9 million were owed to Mauritius and have been canceled since. The remainder was owed to Arab Bank for Economic Development in Africa (BADEA) and OPEC Fund for Development. The authorities are committed to clearing these arrears and have a credible plan for their clearance.

8. Comoros' domestic arrears stock remains highly uncertain and was tentatively estimated KMF 9.1 billion (1.8 percent of GDP) at end-2018. Waiting for a comprehensive audit of the domestic arrears, staff included this amount in the contingent liability test.

A. Macroeconomic Forecasts

9. The medium to long-term macroeconomic assumptions underlying the present DSA incorporate higher GDP and growth numbers, higher services export and lower revenues. The baseline scenario includes April 2019 WEO assumptions and the latest available information on Comoros' debt.

- **Revised GDP series.** In 2018, Comoros' statistical authorities released the 1993 national accounts standards (SNA1993) with help from the IMF and the WB. The level of GDP was revised up by more than 70 percent, and annual growth was revised up by 1.1 percentage points, over 2007-16. Consequently, the public and publicly guaranteed external debt to GDP ratio has declined by 12.6 percentage points to 18.9 percent of GDP in 2018. These revisions impacted also the levels of public and private investment. Over the medium term, growth is expected to accelerate from 3 percent in 2018 to 3.3 percent over the medium and 3.5 percent over the longer term (a rate of growth slightly higher than in the DSA prepared for the 2018 Article IV consultation). Cyclone Kenneth, which hit Comoros in late April 2019, will bring some volatility around this growth path in 2019 and 2020.

- **Revised inflation numbers.** Inflation rate was revised downward by 0.6 percentage points to 1.6 percent between 2012 and 2018. In the short term, inflation rate is expected to reach 3.2 percent in 2019 due to the supply shocks caused by Cyclone Kenneth, 1.4 percent in 2020, and 2 percent over the medium-to long-term. The level of long-term inflation rate remains similar to the previous DSA's.

- **External sector.** The current account deficit (CAD) remains marked by a large goods and services deficit, partially balanced by public and private transfers. Exports of services projections were revised up in line with stronger performance of travelling and tourism in 2017-18. The CAD is projected to peak at 7.7 percent of GDP in 2019, compared to 3.8 percent of GDP in 2018, reflecting in part the impact of the cyclone. Exports of goods will be affected for some years by the loss of part of agricultural plants as a result of the

cyclone, and imports of building and food items will also be higher. The goods and services deficit will be financed through an increase of remittances, aid, and public borrowing.

- **Fiscal.** In 2019, the overall fiscal deficit on a cash basis is expected to widen to 3.3 percent of GDP (from 0.4 percent of GDP in 2018), including grants. The 2019 fiscal situation will be marked by an increase of current and capital spending for reconstruction post-Cyclone Kenneth, which will be partially balanced by an increase of grants. Fiscal revenue is expected to decline by 1.8 percentage points to 9.5 percent of GDP in line with adverse impacts of the cyclone. Starting in 2020, the fiscal situation will gradually normalize, with the deficit converging to 2.3 percent of GDP over the long-run.

- **External borrowing.** As before, the DSA assumes that all contracted-but-undisbursed concessional loans will be fully disbursed over the coming years, as planned by the authorities. New borrowings on concessional terms are also projected to rise over the medium term, however the path will be marked by a larger increase over the reconstruction period. The assumption of increased external borrowing to finance the country's development needs plays a key role in the DSA and explains to some extent the upward slope of the debt burden indicators, particularly for debt level indicators. While this assumption is not based on concrete borrowing plans, it reflects the view that under baseline assumptions, Comoros will wish to borrow into the future to finance productive infrastructure investments. Comoros is also expected to continue to benefit from significant grant financing, leaving the grant element of new borrowing at about 45 percent. Projections include a one-time disbursement of 50 percent of the quota in July 2019, of which 33.3 percent is provided by the Rapid Credit Facility and 66.7 percent by the Rapid Financing Instrument. It includes also a non-concessional financing of US$15 million from the TDB in 2019 (see DSA paragraph 6 above).

- **Domestic borrowing.** As in the previous DSA, domestic borrowing is assumed to remain very limited over the medium-term. Domestic borrowing is expected to reach 4 percent of GDP in 2039, reflecting a gradual deepening of domestic financial markets.

10. The debt sustainability framework's newly-added realism tools suggest that the baseline projections are reasonable (Figures 3 and 4). The difference over 2019-20 between the baseline growth projections and growth projections implied by standard fiscal multipliers reflects the impact of cyclone Kenneth: in 2019, supply constraints lower growth despite a fiscal expansion, and thereafter growth rebounds despite gradual fiscal adjustment as supply constraints recede (see Annex I to this staff report). Similarly, the 3-year adjustment in the fiscal primary balance seems credible, as it does not fall in the upper quartile of the distribution of past adjustments.

B. Country Classification and Determination of Stress Test Scenarios

11. Comoros's debt carrying capacity is assessed as medium (Text Table 4). The April WEO update maintains the Composite Indicator (CI) score at 2.97. The import coverage of reserves constitutes the second largest contributor to Comoros' CI. Under the medium category of debt carrying capacity, the relevant indicative thresholds applicable to the public and publicly

guaranteed external debt are 40 percent for the PV of debt-to-GDP ratio, 180 percent for the PV of debt-to-exports ratio, 15 percent for the debt service-to-exports ratio, and 18 percent for the debt service-to-revenue ratio. The benchmark for the PV of total public debt under medium debt carrying capacity is 55 percent.

Text Table 4. Comoros: Calculation and Contribution of the CI Score Components

	Initial Score	WEO April 2018	WEO October 2018	WEO April 2019
CPIA	2.80	1.09	1.08	1.08
Real growth rate (in percent)		0.07	0.07	0.07
Import coverage of reserves (in percent)		2.19	2.17	2.17
Import coverage of reserves^2 (in percent)		-1.16	-1.14	-1.14
Remittances (in percent)		0.31	0.31	0.31
World economic growth (in percent)		0.49	0.48	0.48
Composite Indicator Score	2.80	2.99	2.97	2.97
Debt Carrying Capacity	Weak	Medium	Medium	Medium

Sources: Comorian authorities, World Bank, and IMF staff.

12. Stress tests follow standardized settings. The contingent liability stress test is based on the quantification of potential contingent liabilities discussed above. The standardized stress tests apply the default settings. Comoros remains exposed and vulnerable to natural disaster shocks, such as tropical cyclones, storms and, the eruption of the Karthala volcano. Consequently, Comoros qualifies for the natural disaster scenario. Given limited mitigating factors, the size of the shock is fixed at 10 percent of GDP.

DEBT SUSTAINABILITY

C. External Debt Sustainability

13. This update suggests that Comoros maintains its external debt distress rating at moderate (Figure 1). As in the previous DSA, all debt and debt service indicators remain below their respective thresholds. However, all the debt and debt service indicators exhibit in the baseline an upward trend over the long run, highlighting limited GDP growth, a narrowed base of export, and only very limited progress in creating fiscal space.

14. The results of the stress tests and alternative scenarios exhibit vulnerability to natural disasters and shocks on exports. A shock to exports is identified as the most extreme shock for the PV of debt-to-exports and the debt service-to-exports ratio.[1] It results in a sustained breach of the relevant threshold for the PV of debt-to-exports, and a large deterioration without breach of

[1] The most extreme stress test is defined as the test that yields the highest level of debt on or before the tenth year of the projection period.

the debt service-to-exports ratio. The most extreme shock for the PV of debt-to-GDP ratio is natural disaster, for debt service-to-revenue ratio it is a one-time depreciation shock. These results highlight the critical importance of improving external competitiveness and enlarging the export base. They also highlight Comoros' vulnerability to natural disasters, and the critical role of exchange rate stability under the peg to the Euro.

15. Comoros debt capacity has limited space to absorb additional shocks (Figure 5). The granularity analysis of the external debt suggests that the PV of debt-to-exports ratio shows limited space to absorb shocks, while the remaining indicators have substantial space.

D. Total Public Debt Sustainability

16. Public-sector domestic debt remains minimal in Comoros. Domestic debt is projected to rise gradually from 0.1 percent of GDP in 2019 to 4 percent of GDP in 2039 (see above).

17. There are no breaches of the public debt benchmark under the baseline or adverse scenarios, signaling limited risks (Figure 2). The PV of total PPG debt-to-GDP increases gradually from 13.6 in 2019 to 29 percent in 2039 but remains well below the threshold. Natural disasters are identified as the most extreme shock for both the PV of debt-to-GDP and the PV of debt-to-revenue, and a one-time depreciation constitutes the most extreme shock for the debt service-to-revenue ratio.

Authorities' Views

18. The authorities agree on the importance of maintaining the risk of debt distress at "moderate". They commit to seeking grants and concessional financing and to avoid as much as possible non-concessional borrowing.

E. Risk Rating and Vulnerabilities

19. This DSA update maintains Comoros' external debt distress rating at "moderate" but indicates that the space to absorb shocks is limited. Under current policies, Comoros may well slide into high risk of external debt distress in the next few years as a result of a breach of the PV of debt to exports ratio. In addition, the shocks considered continue to highlight Comoros' vulnerability to a deterioration of export performance, natural disasters and exchange rate instability.

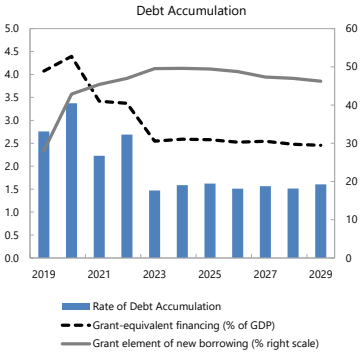
20. These DSA results underscore the need to strengthen both external competitiveness and debt management capacity. Making faster progress on domestic resource mobilization and enlarging the export base are important to improve macroeconomic performance. Also, cautiously seeking for concessional debt contributes not to slide into high risk of external debt distress.

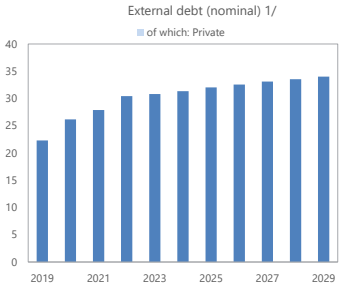
Table 1. Comoros: External Debt Sustainability Framework, Baseline Scenario, 2016-39

(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2016	2017	2018	2019	2020	2021	2022	2023	2024	2029	2039	Historical	Projections
External debt (nominal) 1/	**15.5**	**17.6**	**18.9**	**22.3**	**26.2**	**27.9**	**30.4**	**30.8**	**31.3**	**34.0**	**37.7**	**19.9**	**30.4**
of which: public and publicly guaranteed (PPG)	**15.5**	**17.6**	**18.9**	**22.3**	**26.2**	**27.9**	**30.4**	**30.8**	**31.3**	**34.0**	**37.7**	**19.9**	**30.4**
Change in external debt	2.0	2.1	1.3	3.4	3.9	1.7	2.6	0.4	0.5	0.5	0.3		
Identified net debt-creating flows	**3.4**	**1.0**	**1.8**	**6.8**	**5.9**	**4.7**	**3.2**	**2.7**	**2.5**	**1.3**	**-0.7**	**1.5**	**3.1**
Non-interest current account deficit	**4.2**	**2.1**	**3.9**	**7.6**	**7.2**	**6.0**	**4.5**	**4.1**	**4.2**	**3.1**	**1.3**	**2.9**	**4.6**
Deficit in balance of goods and services	15.6	16.4	17.7	22.9	21.6	20.0	18.1	17.7	17.8	16.7	15.4	**19.0**	**18.5**
Exports	10.5	11.9	13.2	12.8	12.7	12.8	13.0	13.1	13.1	13.4	14.3		
Imports	26.1	28.3	31.0	35.7	34.3	32.8	31.0	30.7	31.0	30.1	29.6		
Net current transfers (negative = inflow)	-10.7	-13.8	-13.2	-14.7	-13.8	-13.4	-12.9	-12.9	-13.0	-13.2	-13.8	**-15.8**	**-13.3**
of which: official	-1.4	-3.5	-1.7	-2.7	-1.9	-1.7	-1.2	-1.2	-1.2	-1.1	-0.9		
Other current account flows (negative = net inflow)	-0.6	-0.5	-0.7	-0.6	-0.6	-0.6	-0.6	-0.6	-0.6	-0.4	-0.3	**-0.3**	**-0.5**
Net FDI (negative = inflow)	**-0.3**	**-0.4**	**-0.6**	**-0.7**	**-0.7**	**-0.7**	**-0.7**	**-0.7**	**-1.0**	**-1.1**	**-1.2**	**-0.9**	**-0.9**
Endogenous debt dynamics 2/	**-0.4**	**-0.7**	**-1.4**	**-0.1**	**-0.7**	**-0.6**	**-0.6**	**-0.7**	**-0.7**	**-0.8**	**-0.8**		
Contribution from nominal interest rate	0.1	0.1	0.1	0.1	0.2	0.3	0.3	0.3	0.3	0.3	0.5		
Contribution from real GDP growth	-0.3	-0.4	-0.5	-0.2	-0.9	-0.9	-0.9	-1.0	-1.0	-1.1	-1.3		
Contribution from price and exchange rate changes	-0.2	-0.3	-1.1	...	...	...	...	...	...	...	...		
Residual 3/	**-1.4**	**1.1**	**-0.6**	**-3.4**	**-2.0**	**-3.0**	**-0.6**	**-2.3**	**-2.0**	**-0.8**	**1.0**	**-3.6**	**-1.7**
of which: exceptional financing	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**10.9**	**13.5**	**15.9**	**17.0**	**18.6**	**18.9**	**19.3**	**21.5**	**25.1**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**82.5**	**105.9**	**125.2**	**133.0**	**143.2**	**144.6**	**147.2**	**160.2**	**176.1**		
PPG debt service-to-exports ratio	**3.0**	**2.8**	**3.2**	**7.5**	**6.2**	**5.8**	**5.4**	**6.6**	**6.3**	**8.0**	**10.6**		
PPG debt service-to-revenue ratio	**4.0**	**3.5**	**3.9**	**10.9**	**8.8**	**8.2**	**7.5**	**9.1**	**8.6**	**10.7**	**14.2**		
Gross external financing need (Million of U.S. dollars)	43.1	22.3	43.7	94.1	93.1	82.2	64.9	65.4	65.4	65.5	59.5		
Key macroeconomic assumptions													
Real GDP growth (in percent)	2.6	3.0	3.0	1.3	4.2	3.6	3.5	3.5	3.5	3.5	3.6	**3.1**	**3.4**
GDP deflator in US dollar terms (change in percent)	1.3	2.1	6.4	-0.3	2.3	2.6	2.6	2.3	2.6	1.9	2.0	**-0.2**	**2.0**
Effective interest rate (percent) 4/	0.5	0.4	0.6	0.7	1.1	1.1	1.1	1.1	1.1	1.1	1.3	**0.5**	**1.1**
Growth of exports of G&S (US dollar terms, in percent)	11.4	18.7	22.0	-2.4	5.6	7.3	7.6	6.9	6.5	6.0	6.6	**8.4**	**5.6**
Growth of imports of G&S (US dollar terms, in percent)	0.2	14.0	20.1	16.3	2.5	1.8	0.3	5.0	6.9	5.3	5.6	**4.1**	**5.2**
Grant element of new public sector borrowing (in percent)	...	...	...	28.1	42.9	45.4	47.0	49.5	49.6	46.2	41.2	**...**	**45.6**
Government revenues (excluding grants, in percent of GDP)	7.9	9.3	11.0	8.8	8.9	9.1	9.3	9.5	9.7	10.1	10.6	**8.6**	**9.6**
Aid flows (in Million of US dollars) 5/	8.5	27.9	12.0	64.3	97.8	74.3	83.0	59.4	64.8	85.6	156.9		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	4.1	4.4	3.4	3.4	2.5	2.6	2.5	2.3	**...**	**3.0**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	53.1	57.3	62.3	58.4	65.3	64.6	60.6	54.0	**...**	**61.1**
Nominal GDP (Million of US dollars)	1,027	1,080	1,184	1,195	1,274	1,355	1,438	1,523	1,616	2,113	3,641		
Nominal dollar GDP growth	3.9	5.2	9.5	1.0	6.6	6.3	6.2	5.9	6.1	5.5	5.7	**2.9**	**5.4**
Memorandum items:													
PV of external debt 7/	...	...	10.9	13.5	15.9	17.0	18.6	18.9	19.3	21.5	25.1		
In percent of exports	...	...	82.5	105.9	125.2	133.0	143.2	144.6	147.2	160.2	176.1		
Total external debt service-to-exports ratio	3.0	2.8	3.2	7.5	6.2	5.8	5.4	6.6	6.3	8.0	10.6		
PV of PPG external debt (in Million of US dollars)			129.3	161.9	202.2	230.6	267.1	288.2	312.4	453.4	915.3		
(PVt-PVt-1)/GDPt-1 (in percent)				2.8	3.4	2.2	2.7	1.5	1.6	1.6	1.7		
Non-interest current account deficit that stabilizes debt ratio	2.2	0.0	2.6	4.2	3.3	4.3	1.9	3.7	3.7	2.6	1.0		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + εα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, ε=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

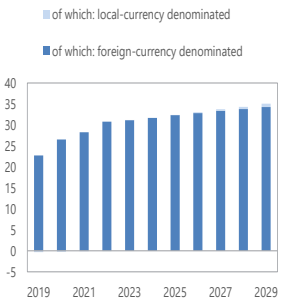
Table 2. Comoros: Public Sector Debt Sustainability Framework, Baseline Scenario, 2016-39

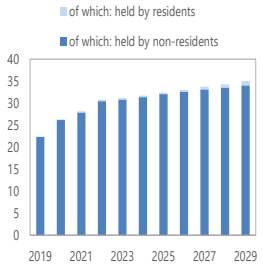
(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2016	2017	2018	2019	2020	2021	2022	2023	2024	2029	2039	Historical	Projections
Public sector debt 1/	**15.6**	**17.8**	**18.9**	**22.4**	**26.3**	**28.2**	**30.7**	**31.2**	**31.7**	**35.1**	**41.7**	**20.5**	**30.8**
of which: external debt	**15.5**	**17.6**	**18.9**	**22.3**	**26.2**	**27.9**	**30.4**	**30.8**	**31.3**	**34.0**	**37.7**	**19.9**	**30.4**
Change in public sector debt	1.2	2.2	1.1	3.5	3.9	1.8	2.6	0.4	0.5	0.8	0.6		
Identified debt-creating flows	**7.8**	**3.7**	**3.6**	**9.5**	**7.4**	**6.4**	**5.7**	**5.4**	**5.6**	**4.7**	**4.6**	**2.0**	**5.9**
Primary deficit	**8.3**	**4.5**	**5.0**	**9.7**	**8.5**	**7.7**	**7.0**	**6.9**	**7.0**	**6.2**	**6.2**	**2.6**	**7.1**
Revenue and grants	8.7	11.9	12.0	11.5	10.9	10.8	10.6	10.8	10.9	11.1	11.6	**12.5**	**11.0**
of which: grants	0.8	2.6	1.0	2.7	1.9	1.7	1.2	1.2	1.2	1.1	0.9		
Primary (noninterest) expenditure	17.0	16.5	16.9	21.2	19.4	18.5	17.6	17.6	17.9	17.3	17.8	**15.1**	**18.1**
Automatic debt dynamics	**-0.5**	**-0.8**	**-1.3**	**-0.2**	**-1.1**	**-1.3**	**-1.3**	**-1.4**	**-1.5**	**-1.4**	**-1.6**		
Contribution from interest rate/growth differential	-0.5	-0.4	-0.6	-0.2	-0.9	-1.1	-1.1	-1.3	-1.5	-1.4	-1.6		
of which: contribution from average real interest rate	-0.2	0.0	0.0	0.1	0.0	-0.2	-0.2	-0.3	-0.4	-0.3	-0.2		
of which: contribution from real GDP growth	-0.4	-0.4	-0.5	-0.2	-0.9	-0.9	-1.0	-1.0	-1.0	-1.2	-1.4		
Contribution from real exchange rate depreciation	0.0	-0.4	-0.8	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**-6.6**	**-1.5**	**-2.5**	**-6.0**	**-3.7**	**-4.7**	**-3.3**	**-5.1**	**-5.0**	**-4.0**	**-4.0**	**-4.1**	**-4.5**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**10.9**	**13.6**	**16.0**	**17.2**	**18.8**	**19.2**	**19.6**	**22.5**	**29.0**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**91.4**	**118.5**	**147.0**	**159.1**	**178.5**	**178.7**	**180.6**	**201.5**	**251.2**		
Debt service-to-revenue and grants ratio 3/	**4.5**	**4.1**	**3.6**	**8.4**	**7.8**	**7.9**	**8.1**	**9.7**	**9.5**	**13.6**	**23.6**		
Gross financing need 4/	8.7	5.0	5.4	10.7	9.4	8.5	7.9	7.9	8.1	7.7	8.9		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	2.6	3.0	3.0	1.3	4.2	3.6	3.5	3.5	3.5	3.5	3.6	**3.1**	**3.4**
Average nominal interest rate on external debt (in percent)	0.5	0.4	0.6	0.7	1.1	1.1	1.1	1.1	1.1	1.1	1.3	**0.5**	**1.1**
Average real interest rate on domestic debt (in percent)	-1.5	-0.1	2.3	-5.9	1.8	1.0	1.1	1.1	1.1	1.0	1.0	**-1.3**	**0.5**
Real exchange rate depreciation (in percent, + indicates depreciation)	0.1	-2.8	-4.5	...	...	...	...	...	...	...	...	**2.3**	**...**
Inflation rate (GDP deflator, in percent)	1.6	0.1	-2.2	7.2	1.2	2.0	1.9	1.9	1.9	1.9	2.0	**1.4**	**2.3**
Growth of real primary spending (deflated by GDP deflator, in percent)	-2.0	-0.3	5.8	26.6	-4.4	-1.2	-1.6	3.9	5.1	4.2	4.7	**4.8**	**3.8**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	7.1	2.3	3.9	6.2	4.6	5.8	4.4	6.4	6.5	5.4	5.6	**4.4**	**5.6**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government plus social security, government-guaranteed debt. Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

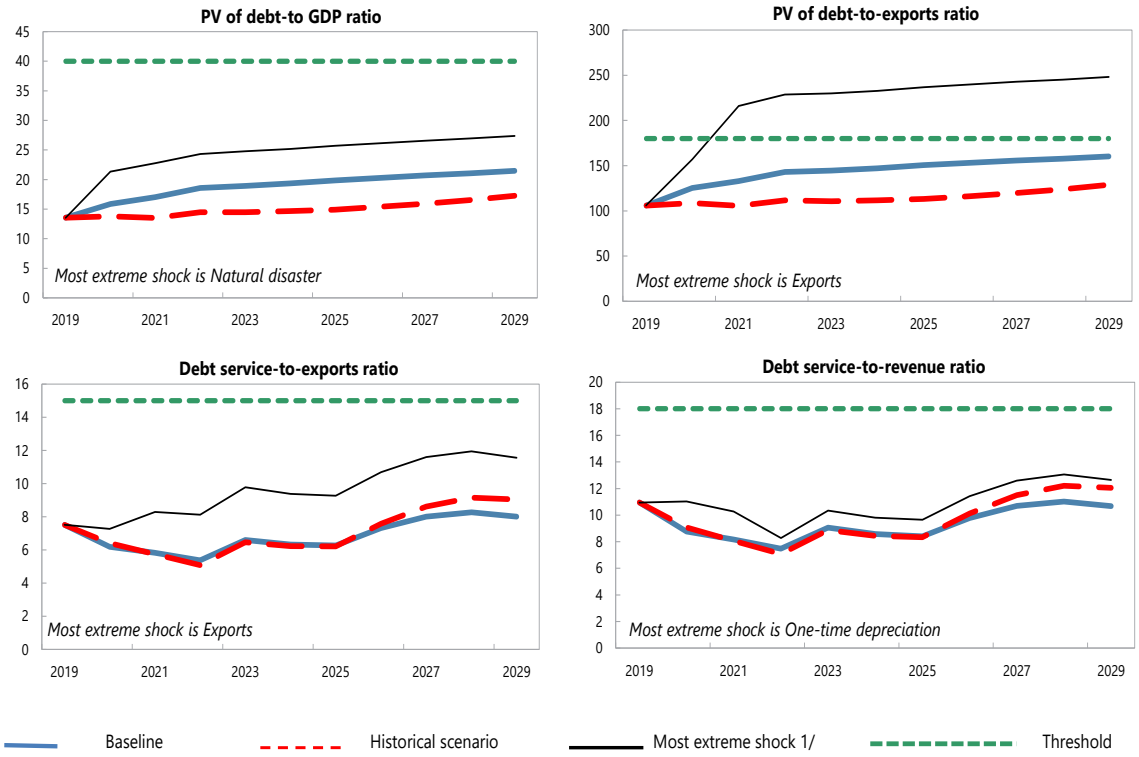
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Comoros: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2019-29 1/



Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	Yes	
Natural Disasters	No	No
Commodity Prices 2/	n.a.	n.a.
Market Financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.4%	1.4%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	29	29
Avg. grace period	9	9

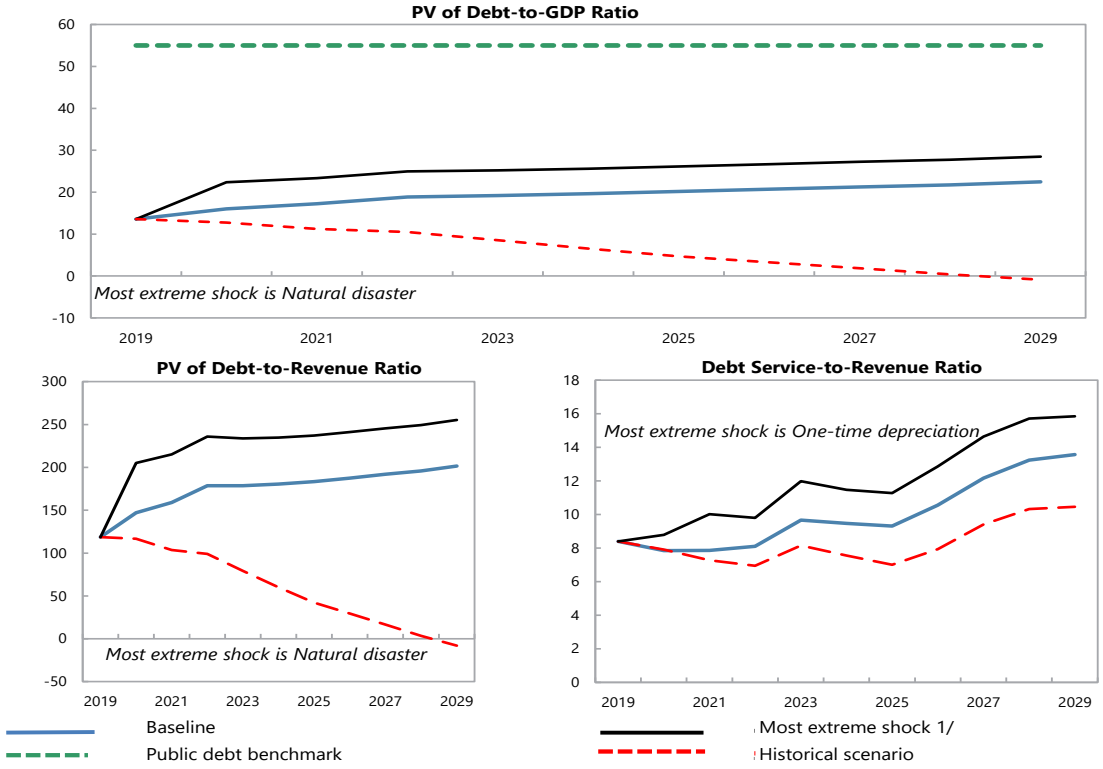
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2029. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Comoros: Indicators of Public Debt Under Alternative Scenarios, 2019-29



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	91%	91%
Domestic medium and long-term	3%	3%
Domestic short-term	6%	6%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	1.4%	1.4%
Avg. maturity (incl. grace period)	29	29
Avg. grace period	9	9
Domestic MLT debt		
Avg. real interest rate on new borrowing	0.7%	0.7%
Avg. maturity (incl. grace period)	3	3
Avg. grace period	2	2
Domestic short-term debt		
Avg. real interest rate	0%	0.0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2029. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Comoros: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2019-29 (in percent)

	Projections 1/										
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
PV of debt-to GDP ratio											
Baseline	13.5	15.9	17.0	18.6	18.9	19.3	19.9	20.3	20.7	21.0	21.5
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	13.5	13.8	13.5	14.5	14.5	14.7	14.9	15.4	15.9	16.5	17.3
B. Bound Tests											
B1. Real GDP growth	13.5	16.4	18.0	19.7	20.0	20.5	21.0	21.5	21.9	22.3	22.7
B2. Primary balance	13.5	17.9	21.2	22.8	23.2	23.5	24.0	24.4	24.7	25.0	25.4
B3. Exports	13.5	17.1	20.4	21.9	22.2	22.6	23.1	23.4	23.8	24.2	24.6
B4. Other flows 3/	13.5	17.9	21.0	22.5	22.8	23.1	23.6	23.9	24.3	24.6	25.0
B5. One-time 30 percent nominal depreciation	13.5	20.0	16.9	18.9	19.4	20.0	20.8	21.3	21.9	22.4	23.0
B6. Combination of B1-B5	13.5	19.3	20.6	22.3	22.6	23.0	23.6	24.0	24.4	24.8	25.2
C. Tailored Tests											
C1. Combined contingent liabilities	13.5	21.4	22.7	24.1	24.5	24.8	25.3	25.6	26.0	26.3	26.6
C2. Natural disaster	13.5	21.4	22.8	24.3	24.8	25.2	25.7	26.1	26.6	26.9	27.4
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	40	40	40	40	40	40	40	40	40	40	40
PV of debt-to-exports ratio											
Baseline	105.9	125.2	133.0	143.2	144.6	147.2	150.6	153.1	155.7	157.7	160.2
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	105.9	108.7	105.6	111.7	110.6	111.7	113.1	116.2	119.8	123.7	128.8
B. Bound Tests											
B1. Real GDP growth	105.9	125.2	133.0	143.2	144.6	147.2	150.6	153.1	155.7	157.7	160.2
B2. Primary balance	105.9	141.6	166.0	175.9	176.9	179.2	**182.1**	**184.1**	**186.2**	**187.7**	**189.6**
B3. Exports	105.9	157.1	**215.9**	**228.7**	**229.8**	**232.7**	**236.8**	**239.8**	**242.9**	**245.2**	**248.1**
B4. Other flows 3/	105.9	141.2	164.3	173.3	173.9	175.9	178.7	**180.7**	**182.9**	**184.4**	**186.4**
B5. One-time 30 percent nominal depreciation	105.9	125.2	104.7	115.8	118.0	121.2	125.1	128.0	131.1	133.5	136.4
B6. Combination of B1-B5	105.9	152.9	147.1	**182.8**	**184.0**	**186.7**	**190.4**	**193.1**	**195.9**	**198.0**	**200.6**
C. Tailored Tests											
C1. Combined contingent liabilities	105.9	168.6	177.3	**186.0**	**187.4**	**189.2**	**191.8**	**193.6**	**195.6**	**196.9**	**198.6**
C2. Natural disaster	105.9	171.8	**181.3**	**191.0**	**192.9**	**195.4**	**198.7**	**201.2**	**203.9**	**205.8**	**208.2**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	180	180	180	180	180	180	180	180	180	180	180
Debt service-to-exports ratio											
Baseline	7.5	6.2	5.8	5.4	6.6	6.3	6.3	7.3	8.0	8.3	8.0
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	7.5	6.4	5.7	5.1	6.5	6.2	6.2	7.6	8.6	9.2	9.0
B. Bound Tests											
B1. Real GDP growth	7.5	6.2	5.8	5.4	6.6	6.3	6.3	7.3	8.0	8.3	8.0
B2. Primary balance	7.5	6.2	6.2	6.1	7.3	7.0	6.9	8.0	8.6	8.9	8.6
B3. Exports	7.5	7.3	8.3	8.1	9.8	9.4	9.3	10.7	11.6	11.9	11.6
B4. Other flows 3/	7.5	6.2	6.2	6.1	7.3	7.0	6.9	7.9	8.6	8.8	8.5
B5. One-time 30 percent nominal depreciation	7.5	6.2	5.8	4.7	6.0	5.7	5.7	6.8	7.5	7.8	7.5
B6. Combination of B1-B5	7.5	6.7	7.2	6.6	8.0	7.7	7.6	8.8	9.6	9.9	9.6
C. Tailored Tests											
C1. Combined contingent liabilities	7.5	6.2	6.8	6.3	7.5	7.2	7.1	8.2	8.8	9.0	8.7
C2. Natural disaster	7.5	6.4	7.0	6.5	7.8	7.5	7.4	8.5	9.1	9.4	9.1
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	15	15	15	15	15	15	15	15	15	15	15
Debt service-to-revenue ratio											
Baseline	10.9	8.8	8.2	7.5	9.1	8.6	8.4	9.8	10.7	11.0	10.7
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	10.9	9.1	8.1	7.1	8.9	8.4	8.3	10.1	11.5	12.2	12.0
B. Bound Tests											
B1. Real GDP growth	10.9	9.1	8.6	7.9	9.6	9.1	8.9	10.4	11.3	11.7	11.3
B2. Primary balance	10.9	8.8	8.7	8.5	10.0	9.5	9.3	10.7	11.5	11.8	11.4
B3. Exports	10.9	8.8	8.6	8.4	9.9	9.4	9.2	10.6	11.4	11.8	11.4
B4. Other flows 3/	10.9	8.8	8.7	8.4	10.0	9.4	9.2	10.6	11.4	11.7	11.3
B5. One-time 30 percent nominal depreciation	10.9	11.0	10.3	8.3	10.3	9.8	9.7	11.4	12.6	13.1	12.6
B6. Combination of B1-B5	10.9	9.4	9.5	8.7	10.4	9.8	9.6	11.1	12.1	12.4	12.0
C. Tailored Tests											
C1. Combined contingent liabilities	10.9	8.8	9.5	8.8	10.3	9.8	9.6	10.9	11.8	12.0	11.6
C2. Natural disaster	10.9	8.8	9.5	8.8	10.3	9.8	9.6	10.9	11.8	12.1	11.7
C3. Commodity price	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
C4. Market Financing	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
Threshold	18	18	18	18	18	18	18	18	18	18	18

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Comoros: Sensitivity Analysis for Key Indicators of Public Debt, 2019-29
(in percent)

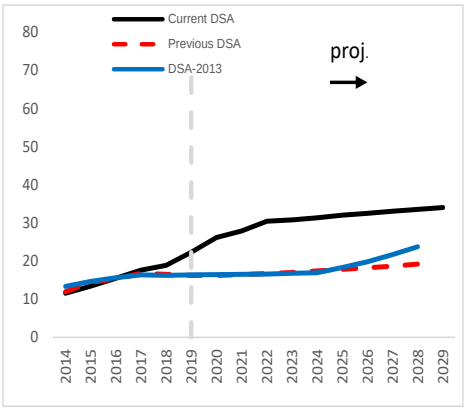
	Projections 1/										
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
PV of Debt-to-GDP Ratio											
Baseline	13.6	16.0	17.2	18.8	19.2	19.6	20.2	20.7	21.3	21.7	22.5
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	14	13	11	10	9	7	5	3	2	0	-1
B. Bound Tests											
B1. Real GDP growth	14	17	19	21	22	22	23	24	25	26	27
B2. Primary balance	14	18	22	23	24	24	24	25	25	26	26
B3. Exports	14	17	20	22	22	22	23	23	24	24	25
B4. Other flows 3/	14	18	21	23	23	23	24	24	25	25	26
B5. One-time 30 percent nominal depreciation	14	19	19	19	19	18	18	18	17	17	17
B6. Combination of B1-B5	14	18	19	19	19	19	20	20	21	21	22
C. Tailored Tests											
C1. Combined contingent liabilities	14	22	23	25	25	25	26	26	27	27	28
C2. Natural disaster	14	22	23	25	25	26	26	27	27	28	28
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Public debt benchmark	55	55	55	55	55	55	55	55	55	55	55
PV of Debt-to-Revenue Ratio											
Baseline	118.5	147.0	159.1	178.5	178.7	180.6	183.3	187.4	191.9	195.7	201.5
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	119	117	103	99	79	60	42	30	17	3	-8
B. Bound Tests											
B1. Real GDP growth	119	153	172	196	199	204	209	216	224	230	239
B2. Primary balance	119	169	203	222	220	220	222	225	229	232	238
B3. Exports	119	157	187	207	206	207	209	213	217	220	225
B4. Other flows 3/	119	166	196	215	214	215	217	220	224	228	233
B5. One-time 30 percent nominal depreciation	119	175	176	183	175	168	164	161	158	155	154
B6. Combination of B1-B5	119	164	177	176	175	177	179	183	188	192	197
C. Tailored Tests											
C1. Combined contingent liabilities	119	206	215	235	232	232	233	237	240	243	248
C2. Natural disaster	119	205	215	236	234	235	237	241	246	249	255
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	8.4	7.8	7.9	8.1	9.7	9.5	9.3	10.6	12.2	13.2	13.6
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	8	8	7	7	8	8	7	8	9	10	10
B. Bound Tests											
B1. Real GDP growth	8	8	8	9	11	11	11	12	14	15	15
B2. Primary balance	8	8	10	11	12	11	10	11	13	14	14
B3. Exports	8	8	8	9	10	10	10	11	13	14	14
B4. Other flows 3/	8	8	8	9	10	10	10	11	13	14	14
B5. One-time 30 percent nominal depreciation	8	9	10	10	12	11	11	13	15	16	16
B6. Combination of B1-B5	8	8	8	9	10	10	9	10	12	13	13
C. Tailored Tests											
C1. Combined contingent liabilities	8	8	14	10	13	11	10	12	13	14	15
C2. Natural disaster	8	8	14	10	14	11	11	12	14	15	15
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.

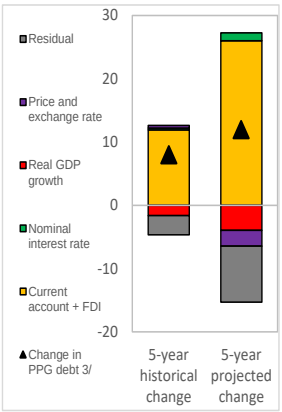
Figure 3. Comoros: Drivers of Debt Dynamics – Baseline Scenario

External debt

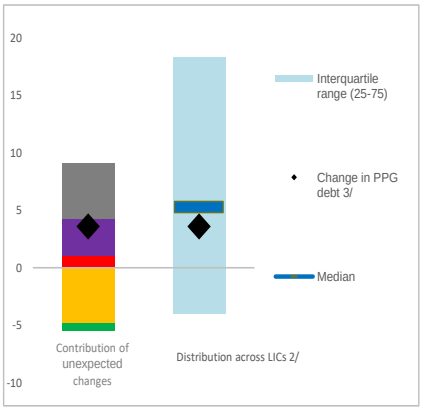
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

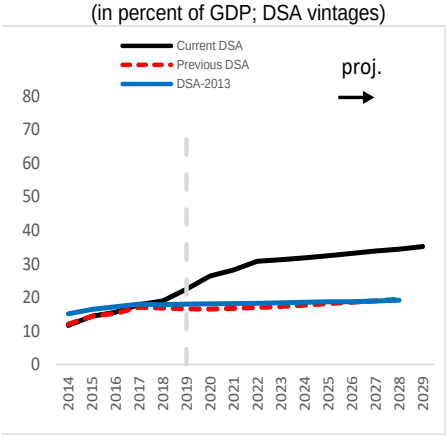


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

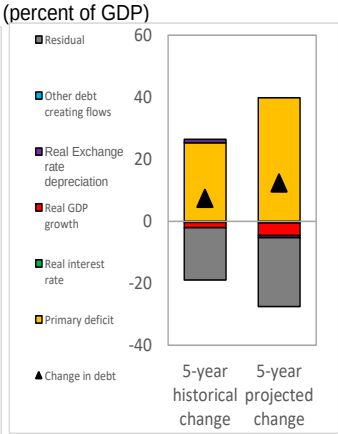


Public debt

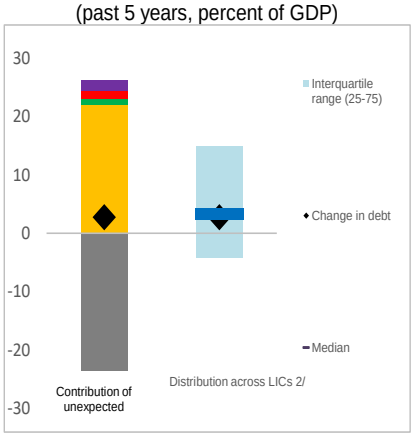
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



1/ Difference between anticipated and actual contributions on debt ratios.

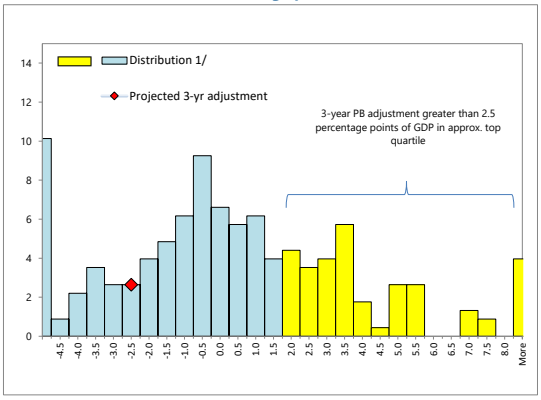
2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

Sources: Comorian authorities, and IMF staff estimates and projections.

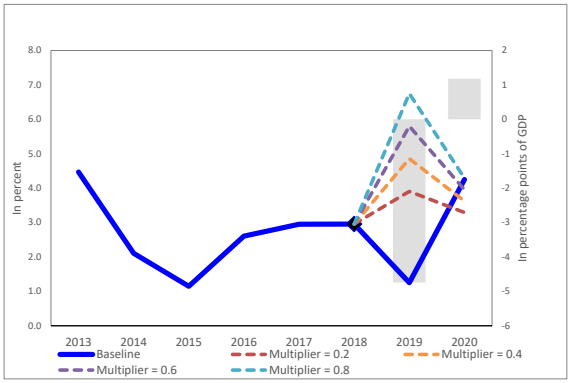
Figure 4. Comoros: Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



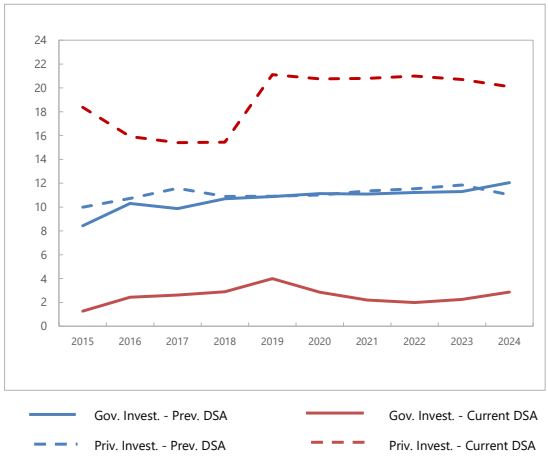
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/

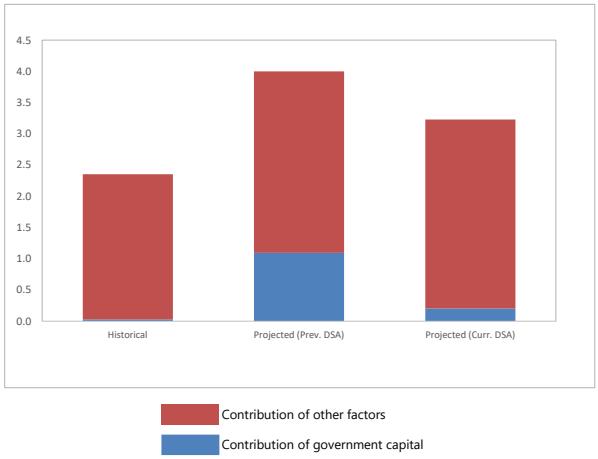


1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(% of GDP)



Contribution to Real GDP growth
(percent, 5-year average)



Sources: Comorian authorities, and IMF staff estimates and projections.

Figure 5. Comoros: Qualification of the Moderate Category, 2019-29 1/

Sources: Country authorities; and staff estimates and projections.

1/ For the PV debt/GDP and PV debt/exports thresholds, x is 20 percent and y is 40 percent. For debt service/Exports and debt service/revenue thresholds, x is 12 percent and y is 35 percent.